ProForma-20Q / Forma: Release Documentation

This is the release documentation for the ProForma-20Q benchmark and the Forma model, and is distributed with the released repositories. Section A documents the data pipeline, from the raw Compustat pull through standardization to the 78-item universe and the identities that link its items. Section B records Forma's training and implementation details and the exact specification of every competitor. Section C documents the LLM benchmark protocol and reproduces the elicitation prompts verbatim. Every number in the paper and in this document is produced by the released pipeline. The one input that does not ship with it is the underlying Compustat panel, which is licensed and cannot be redistributed; the release instead rebuilds it from a WRDS connection, verified against published checksums.

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A Data

This section documents the data underlying the benchmark: the pipeline from the raw Compustat pull to the modeled firm-quarter origins (§A.1), the standardization that maps reported accounting values into model space (§A.2), the 78-item target universe (§A.3), its reporting availability (§A.4), and the accounting identities that link the items (§A.5).

Throughout, a *forecast origin* is a firm-quarter (f, t) from which a model forecasts quarters $t{+}1, \ldots, t{+}20$ using the 12 quarters through the origin $(t{-}11, \ldots, t)$. The benchmark provides the same panel in two views: the *tuple view*, in which each firm-origin is a variable-length set of (relative quarter, item, value) tuples, consumed by FORMA and the LLM protocol, and the *tabular view*, a fixed-width matrix of recent levels and year-over-year changes per item, consumed by the tabular baselines (§B.2).

A.1 Sample formation and splits

Table 1 gives the sample waterfall; the paragraphs below document each stage in order. The scale deflator that defines the final filter is specified in §A.2.

Table 1: Sample formation waterfall. Row 1 is the raw `comp.fundq` pull and each subsequent row applies one filter or transform to the row above. The last waterfall row is the deflator-valid firm-quarter panel whose counts the main paper quotes (dropping the 643 deflator-valid firm-quarters dated 1970, which serve only as history); the rows below the rule partition it by split. Origin *eligibility* additionally requires four quarters of history through the origin (see text).

Stage	Firm-quarters
Raw Compustat quarterly rows (1970–2024 pull)	2,001,259
Distinct firm-quarters (after deduplication)	1,998,293
After financial-sector exclusion (SIC 6000–6999)	1,798,509
With a valid scale deflator	1,342,494
Deflator-valid firm-quarters (1971–2024 window)	1,341,851
Train (1971–2001)	695,257
Validation (2002–2009)	251,059
Test (2010–2024)	395,535

Download and universe. The raw panel is Compustat Fundamentals Quarterly (`comp.fundq`) via WRDS, restricted to `indfmt=INDL`, `datafmt=STD`, `consol=C`, `popsrc=D`, with `datadate` in 1970–2024. Historical SIC codes (`sich`) come from `comp.co_industry`, attached to each firm-quarter as the most recent industry record at or before the statement date. The financial-sector filter drops firm-quarters with `sich` $\in [6000, 6999]$; firm-quarters with *missing* SIC are retained and map to the Unknown industry bucket.

Deduplication and quarter alignment. Firm-quarters are indexed by *calendar* quarter: each row's quarter is the calendar quarter-end of its `datadate`. When multiple rows share a (firm, calendar-quarter) pair (e.g. around fiscal-year changes), rows are sorted by fiscal year and quarter and the last record is kept. Year-to-date accumulation, by contrast, follows the *fiscal* calendar (next paragraph), so flow conversion is unaffected by fiscal/calendar offsets.

Year-to-date conversion. Compustat reports the 20 cash-flow items year-to-date. Each is converted to a quarterly flow within the firm–fiscal-year group: fiscal Q1 values pass through unchanged, and fiscal

Q2–Q4 values are differenced against the immediately preceding fiscal quarter's year-to-date value. If that preceding fiscal quarter is not in the file, the quarterly flow is left missing rather than imputed; no differencing ever crosses a fiscal-year boundary.

Derived items. The six derived items are computed after year-to-date conversion, exactly as defined in Table 2: gross profit (gpq $=$ revtq $-$ cogsq), free cash flow (fcfq $=$ oancfq $-$ capxq), working capital (wcapq $=$ actq $-$ lctq), and the three carve-outs aoq_ex_intanq, loq_ex_dr, xsgaq_ex_rd, in which the subtracted component (intanq, drltq, xrdq) is treated as zero when unreported.

Origin eligibility and splits. A firm-quarter is an eligible forecast origin if it has a valid deflator and at least four quarters of history through the origin. The history requirement is enforced per firm at load time, so a firm's first three panel quarters cannot serve as origins; the tabular build instead drops the panel's first three calendar quarters globally. The two rules differ only at firm entry, and scored comparisons are unaffected because every comparison is made on the common sample of cells all compared models predict. The counts in Table 1 are the deflator-valid firm-quarter panel *before* this final gate; eligible-origin counts are smaller and depend on a consumer's additional requirements (e.g. the LLM protocol's test pool of 370,934 origins, §C.1). Every input window spans the trailing 12 quarters, with unreported (item, quarter) cells contributing no tuple. Splits are by the calendar year of the origin quarter: 1971–2001 train, 2002–2009 validation, 2010–2024 test. Split boundaries also purge targets: a train (validation) example retains only target quarters dated on or before 2001Q4 (2009Q4) — later targets are set to missing in the tabular view and are absent from the split's tuple file — so origins near a boundary contribute only their realized short horizons and no target crosses a split boundary. Hyperparameter and model-selection decisions for every model use the validation split only, whose targets end in 2009Q4; final models are refit on train+validation with all choices frozen (§B.1, §B.2), so nothing trained or selected before test conditions on an outcome after 2009Q4.

A.2 Standardization

Reported values span orders of magnitude across firms and are heavy-tailed within firm. The pipeline therefore maps every accounting value into a standardized space before it reaches any model, in four steps: deflation by firm size, an asinh transform, cross-sectional standardization, and clipping. We call the full mapping from an original value v to a model value x *standardization*, and its inverse *de-standardization*. All standardization parameters are fixed at the forecast origin. The same procedure serves the tuple view and the tabular view, and forecasts are evaluated in the standardized space.

Scale deflator. Let $v_{f,q,\mathrm{id}}$ denote the reported value of item id for firm f at quarter q. The deflator of firm-quarter (f, q) is

$$z_{f,q} \equiv \begin{cases} \left|v_{f,q,\mathtt{ltq}}\right| + \left|v_{f,q,\mathtt{seqq}}\right| + 10^{-3} & \text{when valid,} \\ \left|v_{f,q,\mathtt{atq}}\right| + 10^{-3} & \text{otherwise,} \end{cases}$$

i.e. total liabilities plus total stockholders' equity, which by the balance identity (§A.5) equals total assets less noncontrolling interests, up to sign conventions. The primary definition is valid when both components are reported and the sum is finite and positive; the total-assets fallback rescues 7,364 firm-quarters, and firm-quarters with no valid deflator under either definition are dropped from the panel (456,015 firm-quarters; Table 1). We call the surviving firm-quarters *scale-valid*. For an origin (f, t), every model input and target — at every lead and lag in the window — is deflated by the origin deflator $z_{f,t}$; deflators at other quarters enter only the normalization statistics below.

Scaling and transform. For a firm f with forecast origin t, the scaled value of an accounting tuple (h, id, v) — item id reporting value v at relative quarter h, i.e. absolute quarter $t + h$ — is

$$v^{\mathrm{scaled}}_{h,\mathrm{id}}(f,t) \equiv \operatorname{asinh}\left(k_{\mathrm{id}}\,\frac{v}{z_{f,t}}\right),$$

so the entire example, leads and lags alike, is expressed on one origin-fixed scale. The account-specific constant k_{id} is estimated once on training-period observations (quarters through 2001Q4) of the lagged-ratio pool defined for σ below, and is then held fixed: it is the value on a 250-point logarithmic grid over $[10^{-2}, 10^{3}]$ that brings the kurtosis of the pooled transformed values as close as possible to 3, the Gaussian value.

Cross-sectional standardization. Scaled values are standardized per (item, quarter) with statistics built from trailing cross-sections. Let $\mathcal{F}$ denote the cross-sectional pool of firms. For each calendar quarter q, define the raw per-quarter statistics

$$\mu^{\mathrm{raw}}_{\mathrm{id},q} \equiv \operatorname{mean}_{f'\in\mathcal{F}} \operatorname{asinh}\left(k_{\mathrm{id}}\,\frac{v_{f',q,\mathrm{id}}}{z_{f',q}}\right), \qquad \sigma^{\mathrm{raw}}_{\mathrm{id},q} \equiv \operatorname{std}_{f'\in\mathcal{F},\, r\in\{0,-8\}} \operatorname{asinh}\left(k_{\mathrm{id}}\,\frac{v_{f',q+r,\mathrm{id}}}{z_{f',q-4}}\right).$$

The mean is the contemporaneous cross-section of item id at quarter q, each observation deflated by its own quarter's scale. The standard deviation is computed over the pooled values at quarters q and $q-8$, both deflated by the scale of the intermediate quarter $q-4$ (the lagged-ratio pool). In both constructions a lag indexes positions in the firm's reported series for the item, so reporting gaps collapse. The origin-quarter parameters average the four most recent raw statistics (fewer at the panel start),

$$\mu_{\mathrm{id},t} \equiv \frac{1}{4}\sum_{q=t-3}^{t} \mu^{\mathrm{raw}}_{\mathrm{id},q}, \qquad \sigma_{\mathrm{id},t} \equiv \frac{1}{4}\sum_{q=t-3}^{t} \sigma^{\mathrm{raw}}_{\mathrm{id},q},$$

and the normalized value is

$$v^{\mathrm{normalized}}_{h,\mathrm{id}}(f,t) \equiv \frac{v^{\mathrm{scaled}}_{h,\mathrm{id}}(f,t) - \mu_{\mathrm{id},t}}{\sigma_{\mathrm{id},t} + 10^{-8}},$$

with 10^{-8} a numerical floor. All means and standard deviations use available reported observations only. Every lag in the construction is non-negative and the four-quarter averages are trailing, so $(\mu_{\mathrm{id},t}, \sigma_{\mathrm{id},t})$ depend only on data through quarter t and are updated each quarter; the per-item constants k_{id} are the only standardization parameters frozen at the 2001Q4 estimation cutoff.

Clipping. With $\mathrm{clip}(u, a, b) \equiv \min\{b, \max\{a, u\}\}$, the model value of a tuple (h, id, v) is

$$x_{f,t}(h, \mathrm{id}) \equiv \mathrm{standardize}_{f,t}(h, \mathrm{id}, v) \equiv \mathrm{clip}\left(v^{\mathrm{normalized}}_{h,\mathrm{id}}(f,t), -6, 6\right).$$

The same clipped value serves as a model input when the tuple is visible and as the supervised target when it is masked. In the tuple view the clip is applied at load time; in the tabular view it is applied at build time, to levels and targets at 6 and to year-over-year difference features, which are computed from the unclipped standardized levels, at $6\sqrt{2}$ (the ± 6 bound scaled by $\sqrt{2}$, the standard deviation of the difference of two independent unit-variance values). Because clipping is not invertible, an original value whose standardized value lies at either boundary cannot generally be recovered uniquely.

The scale token. In the tuple view the deflator itself enters as one additional input tuple per origin, under a dedicated scale account at $h = 0$, so models observe firm size. Its value is $z_{f,t}$, standardized like any other account through an asinh transform with its own constant k_{scale} and per-quarter statistics $(\mu_{\text{scale},t}, \sigma_{\text{scale},t})$ estimated on the cross-section of contemporaneous deflator values; the deflator is not divided by itself. The scale account is never queried at future horizons and receives no future-grid slots (§B.1); like any historical tuple, it can be masked for reconstruction during training. In the tabular view the deflator enters as one feature on the same standardized basis.

De-standardization. Given a predicted model value $\widehat{x}_{f,t}(h, \text{id})$, the prediction in original accounting units is

$$\widehat{v}_{f,t+h,\text{id}} \equiv z_{f,t} \frac{\sinh(\widehat{x}_{f,t}(h, \text{id})\, \sigma_{\text{id},t} + \mu_{\text{id},t})}{k_{\text{id}}}$$

(numerical floors of order 10^{-8} omitted). De-standardization uses the deflator and normalization parameters fixed at the forecast origin t, never the realization quarter $t + h$. The map is strictly increasing in $\widehat{x}$, so it carries quantiles of a predictive distribution in model space to quantiles in accounting units, a property the Chronos-2 comparison relies on (§B.2).

A.3 The ProForma-20Q variable universe

ProForma-20Q targets the 78 quarterly Compustat line items in Table 2: 34 balance-sheet, 18 income-statement, 20 cash-flow, and 6 derived items, jointly at horizons $h = 1, \ldots, 20$. The same items serve as both conditioning inputs (reported history) and prediction targets (future quarters); ProForma-20Q additionally conditions on a Fama–French-48 industry category. Cash-flow items are converted from year-to-date to quarterly flows (§A.1); derived items are exact functions of raw items (definitions in the table). Mnemonics are Compustat quarterly field names.

Table 2: The 78-item ProForma-20Q target universe, ordered alphabetically by Compustat mnemonic: description, statement (St: BS balance sheet, IS income statement, CF cash flow, Der derived), and reporting availability. *% raw* is the fraction of all 1,998,293 Compustat quarterly firm-quarters (all sectors and years, no scale-validity requirement) that report the item after year-to-date→quarterly conversion, i.e. the reporting rate before any sample filter. The split columns are that same fraction restricted to the scale-valid firm-quarters of each split: train 1971–2001 (N=695,257), validation 2002–2009 (N=251,059), test 2010–2024 (N=395,535). Items near-zero in the training era (e.g. `stkcoq`, `txbcofq`, `drcq`, `acomincq`) are modern disclosure fields absent from early filings; FORMA represents them as missing rather than imputing them.

Mnemonic	Description	St	% raw	% Tr	% Val	% Te
acomincq	Accumulated other comprehensive income	BS	37.6	1.7	92.6	97.8
acoq	Current assets, other	BS	70.5	92.4	96.8	99.6
actq	Current assets, total	BS	64.2	89.9	94.3	97.5
ancq	Non-current assets, total	BS	61.5	85.6	90.8	93.8
aoq	Assets, other	BS	76.0	98.4	99.8	99.9
aoq_ex_intanq	Other assets excl. intangibles	Der	76.0	98.4	99.8	99.9
apq	Accounts payable	BS	74.9	96.9	98.9	99.3
aqcq	Acquisitions	CF	60.1	65.5	89.7	90.0
atq	Assets, total	BS	76.7	100.0	100.0	100.0
capsq	Capital surplus / APIC	BS	73.7	93.7	95.2	94.6
capxq	Capital expenditures	CF	60.9	65.7	91.4	92.5
ceqq	Common equity, total	BS	77.6	98.7	99.5	99.7
cheq	Cash & short-term investments	BS	75.8	97.8	99.8	99.9
cogsq	Cost of goods sold	IS	79.8	97.4	98.8	99.3
cstkq	Common stock	BS	74.4	94.9	95.3	95.8

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Table 2, continued

Mnemonic	Description	St	% raw	% Tr	% Val	% Te
dlcchq	Change in current debt	CF	33.3	33.3	52.1	54.5
dlcq	Debt in current liabilities	BS	73.2	93.8	97.5	97.6
dltisq	Long-term debt issuance	CF	58.9	63.4	87.1	90.5
dltrq	Long-term debt reduction	CF	59.3	63.6	88.5	91.0
dlttq	Long-term debt	BS	77.4	98.6	99.2	99.2
dpactq	Accumulated depreciation	BS	50.1	73.5	58.7	69.7
dpq	Depreciation & amortization	IS	69.3	81.9	92.4	95.8
drcq	Deferred revenue, current	BS	28.9	0.1	68.2	89.4
drltq	Deferred revenue, long-term	BS	30.6	0.1	70.5	92.6
dvq	Cash dividends	CF	61.2	66.7	90.9	92.1
exreq	Exchange rate effect on cash	CF	55.4	50.0	91.8	92.9
fcfq	Free cash flow = oancfq − capxq	Der	54.7	48.9	91.4	92.5
fiaoq	Financing activities, other	CF	55.3	49.9	91.6	92.6
fincfq	Financing cash flow	CF	55.5	50.1	92.0	93.1
fopoq	Funds from operations, other	CF	58.4	60.5	89.1	91.8
gdwlq	Goodwill	BS	38.3	4.2	91.5	97.0
gpq	Gross profit = revtq − cogsq	Der	74.4	92.2	95.7	99.0
ibq	Income before extraordinary items	IS	82.8	99.4	99.3	99.6
intanoq	Intangibles, other	BS	36.9	4.3	88.1	92.7
intanq	Intangibles, total	BS	41.7	8.6	99.0	99.5
invtq	Inventory	BS	74.0	95.1	97.8	98.2
ivacoq	Investing activities, other	CF	55.3	49.9	91.7	92.7
ivchq	Increase in investments	CF	59.0	64.1	88.0	89.4
ivncfq	Investing cash flow	CF	55.5	50.1	92.0	93.1
ivstchq	Short-term investments, change	CF	44.9	46.0	77.8	73.0
lcoq	Current liabilities, other	BS	70.3	91.8	96.5	99.6
lctq	Current liabilities, total	BS	64.7	90.5	94.4	97.6
loq	Liabilities, other	BS	76.0	98.3	99.8	99.9
loq_ex_dr	Other liab. excl. deferred rev.	Der	76.0	98.3	99.8	99.9
ltq	Liabilities, total	BS	76.4	99.1	99.8	99.9
mibtq	Noncontrolling interests (BS)	BS	72.7	91.9	94.3	98.2
miiq	Noncontrolling interest (income)	IS	76.7	89.3	92.2	96.9
niq	Net income	IS	82.7	99.4	99.2	99.5
nopiq	Non-operating income	IS	79.9	98.1	98.8	99.3
oancfq	Operating cash flow	CF	55.5	50.1	92.0	93.1
oiadpq	Operating income after D&A (EBIT)	IS	79.0	96.8	98.2	99.0
oibdpq	Operating income before D&A (EBITDA)	IS	70.3	83.2	92.7	95.7
piq	Pretax income	IS	81.8	98.4	99.2	99.5
ppegtq	Gross PP&E	BS	50.0	73.5	58.6	69.6
ppentq	Net PP&E	BS	74.7	99.2	99.6	99.5
prstkcq	Purchase of common & preferred stock	CF	58.2	63.2	85.0	88.3
pstkq	Preferred stock	BS	77.6	98.8	99.3	99.7
rectq	Receivables, total	BS	73.5	94.3	97.7	97.5
req	Retained earnings	BS	74.2	94.5	95.6	95.0
revtq	Revenue, total	IS	77.3	94.0	96.1	99.2
seqq	Stockholders' equity, total	BS	78.2	99.9	99.8	99.9
sivq	Sale of investments	CF	59.0	64.0	88.3	89.7
spiq	Special items	IS	77.4	90.0	98.7	98.1
sppeq	Sale of PP&E	CF	50.8	53.4	76.3	79.0
sstkq	Sale of common & preferred stock	CF	60.5	65.6	90.3	91.2
stkcoq	Stock compensation expense	IS	29.3	0.1	61.4	82.8
tstkq	Treasury stock	BS	72.8	87.5	98.9	99.0
txbcofq	Excess tax benefit, stock options	CF	29.4	0.0	48.7	92.3
txditcq	Deferred taxes & investment tax credit	BS	67.2	88.5	92.0	93.8
txpq	Income taxes payable	BS	65.3	84.9	89.0	92.9

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Table 2, continued

Mnemonic	Description	St	% raw	% Tr	% Val	% Te
txtq	Income taxes, total	IS	82.0	98.5	99.2	99.5
wcapq	Working capital = actq − lctq	Der	64.0	89.6	94.2	97.5
xidoq	Extraordinary items & disc. ops	IS	82.7	99.4	99.2	99.5
xintq	Interest expense	IS	65.8	79.0	81.7	88.5
xoprq	Operating expenses, total	IS	79.5	97.2	98.5	99.1
xrdq	R&D expense	IS	24.9	22.9	41.6	44.2
xsgaq	SG&A expense	IS	62.8	77.7	83.3	84.9
xsgaq_ex_rd	SG&A excl. R&D	Der	62.8	77.7	83.3	84.9

A.4 Availability and sparsity

Reporting is pervasively incomplete. Table 3 quantifies the joint availability of the 78-item vector across scale-valid firm-quarters. On the test panel the item×firm-quarter matrix is 93.2% dense and the average firm-quarter reports 73 of 78 items, yet only 4.9% report all 78, so complete-case (listwise) deletion, the implicit requirement of a dense tabular design, would discard 95.1% of the test panel. FORMA's tuple-set representation accommodates this incompleteness natively; the tabular baselines require an explicit imputation model.

Table 3: Joint availability of the 78-item vector across scale-valid firm-quarters. *Matrix density* is the fraction of item×firm-quarter cells reported; *items per firm-quarter* summarizes how many of the 78 items a firm reports in a given quarter; *complete-case* firm-quarters report all 78.

	Full sample 1971–2024	Test 2010–2024
Scale-valid firm-quarters	1,341,851	395,535
Distinct firms	33,469	14,136
Matrix density (%)	82.2	93.2
Items per firm-quarter, mean	64.2	72.7
median [p10, p90]	68 [46, 76]	75 [67, 77]
With ≥ 40 items (%)	95.2	99.5
With ≥ 70 items (%)	41.2	85.8
Complete-case (all 78, %)	1.83	4.94
Cell density by statement (%), full / test		
Balance sheet	85.5	95.8
Income statement	87.7	93.4
Cash flow	69.7	88.1
Derived	89.4	95.6

A.5 Accounting identities

The 26 exact linear identities below tie the target items. They define the identity groups used by FORMA's grouped masking objective and the coherence residuals analyzed in the main paper. Each holds in levels at every quarter; contemporaneous subscripts are suppressed. Signs are normalized so all terms are additive (negative-coefficient terms moved to the right-hand side).

Balance sheet (13).

$$\begin{aligned}
\texttt{cheq} + \texttt{invtq} + \texttt{rectq} + \texttt{acoq} + \texttt{ppentq} + \texttt{aoq} &= \texttt{atq} \\
\texttt{actq} + \texttt{ancq} &= \texttt{atq} \\
\texttt{cheq} + \texttt{invtq} + \texttt{rectq} + \texttt{acoq} &= \texttt{actq} \\
\texttt{wcapq} + \texttt{lctq} &= \texttt{actq} \\
\texttt{intanq} + \texttt{aoq_ex_intanq} &= \texttt{aoq} \\
\texttt{gdwlq} + \texttt{intanoq} &= \texttt{intanq} \\
\texttt{ppentq} + \texttt{dpactq} &= \texttt{ppegtq} \\
\texttt{apq} + \texttt{lcoq} + \texttt{dlcq} + \texttt{txpq} + \texttt{dlttq} + \texttt{txditcq} + \texttt{loq} &= \texttt{ltq} \\
\texttt{apq} + \texttt{dlcq} + \texttt{txpq} + \texttt{lcoq} &= \texttt{lctq} \\
\texttt{drltq} + \texttt{loq_ex_dr} &= \texttt{loq} \\
\texttt{cstkq} + \texttt{capsq} + \texttt{req} &= \texttt{tstkq} + \texttt{ceqq} \\
\texttt{seqq} &= \texttt{pstkq} + \texttt{ceqq} \\
\texttt{atq} &= \texttt{ltq} + \texttt{mibtq} + \texttt{seqq}
\end{aligned}$$

Income statement (10).

$$\begin{aligned}
\texttt{gpq} + \texttt{cogsq} &= \texttt{revtq} \\
\texttt{oibdpq} + \texttt{xsgaq} &= \texttt{gpq} \\
\texttt{xrdq} + \texttt{xsgaq_ex_rd} &= \texttt{xsgaq} \\
\texttt{xoprq} &= \texttt{cogsq} + \texttt{xsgaq} \\
\texttt{oiadpq} + \texttt{dpq} &= \texttt{oibdpq} \\
\texttt{piq} + \texttt{xintq} &= \texttt{oiadpq} + \texttt{nopiq} + \texttt{spiq} \\
\texttt{ibq} + \texttt{txtq} + \texttt{miiq} &= \texttt{piq} \\
\texttt{niq} &= \texttt{ibq} + \texttt{xidoq} \\
\texttt{niq} + \texttt{txtq} + \texttt{miiq} &= \texttt{piq} + \texttt{xidoq} \\
\texttt{revtq} + \texttt{nopiq} + \texttt{spiq} + \texttt{xidoq} &= \texttt{cogsq} + \texttt{xsgaq} + \texttt{dpq} + \texttt{xintq} + \texttt{txtq} + \texttt{miiq} + \texttt{niq}
\end{aligned}$$

Cash flow (3).

$$\begin{aligned}
\texttt{sivq} + \texttt{sppeq} + \texttt{ivstchq} + \texttt{ivacoq} &= \texttt{capxq} + \texttt{ivchq} + \texttt{aqcq} + \texttt{ivncfq} \\
\texttt{sstkq} + \texttt{dltisq} + \texttt{dlcchq} + \texttt{fiaoq} + \texttt{txbcofq} &= \texttt{prstkcq} + \texttt{dltrq} + \texttt{dvq} + \texttt{fincfq} \\
\texttt{fcfq} + \texttt{capxq} &= \texttt{oancfq}
\end{aligned}$$

B Models

The main paper specifies FORMA's representation and objective. This section records the implementation detail needed to retrain it from scratch (§B.1) and the exact specification of every competitor (§B.2). The released configuration files are the authoritative source for all values quoted here. Throughout, the *headline table* is the main paper's headline accuracy table, which scores three tracks on standardized values: a squared-error track summarized by change-space R^2 (the R^2 of predicted against realized changes $x_{t+h} - x_t$, so zero corresponds to predicting the sample-average change), an absolute-error track (MAE), and a density track (log score, CRPS, and coverage) for models that produce predictive distributions.

B.1 FORMA: training and implementation

Table 4 collects the canonical hyperparameters.

Table 4: FORMA canonical hyperparameters (identical across the five mixture seeds; only the seed differs).

Architecture	
Encoder layers / d_{model} / heads	4 / 128 / 4
Feed-forward width / dropout	512 / 0.2
Trainable parameters (incl. variance head)	942,210
Optimization	
Optimizer	AdamW (default β, ϵ)
Learning rate	10^{-4}, constant
Weight decay	0.1 (all parameters)
Gradient-norm clip	1.0
Batch size	32 firm-origin sets
Epochs	12 (final-epoch weights used)
Precision	fp32 (`medium` matmul precision)
Seeds	60–64 (equal-weight mixture)
Loss	
Objective	Gaussian β-NLL, $\beta = 0.5$
$\log \sigma^2$ clamp	$[-10, 10]$
Absolute-error-track variant	Laplace β-NLL, $b = \sigma/\sqrt{2}$
Horizon curriculum	
Initial $\rightarrow$ max horizon	4 $\rightarrow$ 20 quarters
Step	$+4$ per epoch after epoch 2
Lookback	12 quarters, fixed
Masking probabilities (per training example)	
Mask entire future block	0.5
else mask each future tuple	0.95
Select historical identity instance	0.1 (mask $k \sim \mathrm{U}\{2, \ldots, N\}$ of its N members)
Mask ungrouped historical tuple	0.05

Embeddings. All embedding components share $d_{\text{model}} = 128$: a learned account-id embedding over the build's account vocabulary, a learned industry embedding over the 48 Fama–French industries plus an Unknown bucket, a fixed (parameter-free) sinusoidal encoding of the quarter offset relative to the origin, a learned value direction w_x that scales with the standardized value, and a learned mask vector substituted for the value component of hidden tuples. The industry token is never masked and carries no prediction target.

Output heads. The final representation of each queried tuple is concatenated with its sinusoidal horizon encoding (width $2d_{\text{model}} = 256$) and passed through two parallel two-layer MLPs ($256 \to 256 \to 1$, GELU): one for the mean μ and one for the log-variance $\log \sigma^2$. The log-variance is clamped to $[-10, 10]$ before exponentiation; there is no other floor on σ.

Loss. With $\beta = 0.5$ the per-tuple Gaussian loss is

$$\ell \;=\; \left[\tfrac{(y-\mu)^2}{2\sigma^2} + \tfrac{1}{2}\log\sigma^2\right] \cdot \lfloor \sigma^2 \rfloor_{\text{sg}}^{\beta},$$

where y is the standardized target value of §A.2, $\lfloor\cdot\rfloor_{\text{sg}}$ denotes stop-gradient (Seitzer et al., 2022), and $\beta = 0$ recovers the standard NLL. The loss averages over tuples that are both masked and observed: synthetic query tokens without realized values (below) and the industry token never enter the loss. The absolute-error-track variant is a separately trained five-seed family, identical except that the head is read as a Laplace scale $b = \sigma/\sqrt{2}$ and trained with the Laplace β-NLL.

Masking procedure. For each training example: (i) with probability 0.5 the entire future block is masked (pure forecasting); otherwise each future tuple is masked independently with probability 0.95, revealing ≈5% of realized future values as inputs (the mechanism that lets a user pin assumed future values and condition the remaining forecasts on them; see the main paper's scenario analysis); (ii) each complete historical identity instance (all member accounts reported in a quarter) is selected with probability 0.1, and a selected instance masks k members with k drawn uniformly from $\{2, \ldots, N\}$; k is never exactly one, so no masked value is recoverable from its own identity; (iii) historical tuples of *accounts* that participate in no complete historical identity instance in the window are masked independently with probability 0.05. An account appearing in a complete instance in any historical quarter is exempt from this singleton masking at all its quarters.

Curriculum. Training runs 12 epochs with the maximum queried horizon following the schedule 4, 4, 8, 12, 16, 20,. . . , 20 (the first expansion comes after the second epoch); the training loader is rebuilt each epoch at the current horizon. The canonical run trains on the merged train and validation panel (1971–2009), like the refit tabular competitors of §B.2: all hyperparameters were fixed beforehand from validation-split experiments, no validation pass runs during training, and final-epoch weights are used. The 12-quarter lookback is fixed for every example; the 4-quarter minimum history only gates which firm-quarters are eligible origins.

Batching. One example is the full tuple set of one firm-origin. No maximum sequence length is imposed and no example is truncated; sets are batched 32 at a time and padded to the longest set in the batch with a boolean key-padding attention mask. Across the full panel, the average origin carries ≈680 observed historical tuples and ≈1,016 realized future targets; the model's token set additionally includes the synthetic query tokens for unrealized (account, horizon) cells (below), the scale token of §A.2, and one industry token.

Ex-ante future grid. Future queries are built ex ante: for each origin, a synthetic query tuple is created for every (account, horizon) pair with the account drawn from the set the firm reports *at the origin quarter* and the horizon running $1, \ldots, 20$, truncated only where the horizon extends past the split's last calendar quarter. That cut is identical for every firm and does not depend on the firm's own survival. Queries whose realization is later observed are loss-eligible targets; queries never realized (e.g. the firm exits) remain in the set as masked tokens but are excluded from the loss and masked at scoring time.

Inference. At test time all future tuples are masked and a single forward pass yields (μ, σ) for every queried (item, horizon) pair jointly, with no iterated roll-forward. Forecasts are produced in fp32; $\sigma = \exp(\frac{1}{2}\log\sigma^2)$ is saved alongside μ. No reconciliation is applied at inference; the ex-post projections studied in the main paper are computed downstream from the saved forecasts.

Mixture. The five seeds are trained independently and combined as an equal-weight mixture. Distributional metrics use the exact mixture: log-scores from the mixture density and CRPS from closed-form mixture expressions.

Compute. Each seed trains in ≈13 hours on a single NVIDIA A100-SXM GPU (≈9 hours on an H100-SXM); the five-seed run cost ≈$124 of on-demand cloud compute. A trained checkpoint is ≈11 MB.

B.2 Competitor specifications

All learned tabular competitors share one protocol. They consume the tabular view: for each of the 78 items, four standardized recent levels and eight standardized year-over-year changes, plus the standardized deflator (§A.2) and 48 industry dummies, with the Unknown bucket as the omitted reference (985 features in all). Missing features are imputed with the cross-sectional latent-factor model of Bryzgalova et al. (2025) with 10 factors, fit one calendar quarter at a time and filling only unobserved cells. Hyperparameters are selected on the 2002–2009 validation split (never test) and final models are refit on train+validation.

Elastic net. One `sklearn` elastic net per (item, horizon) pair (1,560 fits; rows with any missing feature or target are dropped per fit). Grid: $\alpha \in \{0.01, 0.1, 1, 10, 100, 1000\} \times \ell_1$-ratio $\in \{0.01, 0.1, 0.3, 0.5, 0.7, 0.9, 0.99\}$, selected by validation MSE per horizon on a reference item (net income) and shared across items at that horizon; coordinate descent, max 10,000 iterations.

Random forest. One GPU (cuML) regression forest per (item, horizon) pair. Fixed parameters: 25 minimum samples per leaf, 50% bootstrap per tree. Grid (same selection protocol as the elastic net): trees $\in \{50, 100, 200\}$, `max_depth` $\in \{5, 10, 20\}$, `min_samples_split` $\in \{100, 200\}$. The full 1,560-model fit runs in about 44 hours on one GPU (A100-SXM). The forest's random seed was not passed through to cuML in the reported run, so the RF arm is not bitwise reproducible.

Feed-forward networks. Each FFNN is a single multi-output network mapping the 985 features to two 1,560-wide heads (a mean and a log-variance per (item, horizon)), trained with the same masked Gaussian β-NLL ($\beta = 0.5$, $\log\sigma^2$ clamped to $[-10, 10]$) as FORMA; missing targets contribute zero loss rather than dropping the row, and rows are dropped only for residual missing features (rare after imputation). The *linear* variant has no hidden layer (3,076,320 parameters); the *large* variant has GELU hidden layers of widths 1024/1024/512 with dropout 0.1 (4,184,624 parameters), 3.3 and 4.4 times FORMA's 942,210 respectively. AdamW, learning rate 10^{-4}, no weight decay, batch 1024; trained for a fixed 8 (linear) and 2 (large) epochs (the best-validation epochs from a held-out selection run) on train+validation. Like FORMA, each variant is a five-seed (60–64) equal-weight mixture with saved σ, so both enter the density track.

Chained GBM. Our re-implementation of the chained gradient-boosting approach of Geertsema et al. (2026) under the common protocol. LightGBM regressors are organized in a 14-step chain over 28 of the original specification's 29 core items (`iva` is dropped as scarcely populated in the quarterly file); the chain runs revenue first, then costs and working capital, through the capital-structure and tax blocks. At each step the feature matrix is augmented with the chain's upstream items: realized values during training

(teacher forcing), predicted values at inference (cascade), with unreported upstream values entering as zero. Per item: up to 500 trees with early stopping (patience 10) on the validation split, leaves $\in \{15, 31\}$ selected on validation, refit on train+validation at the selected size. Two arms are trained, an L_1 objective (the original specification, absolute-error track) and an L_2 objective (squared-error track). Of the 28 chained items, 25 lie in the 78-item ProForma-20Q universe; the headline table's Geert. column is scored on this 25-item footprint.

Pooled fade/AR(1). For each (item, horizon), one pooled OLS regression of the standardized future value on the standardized origin value, $\hat{x}_{t+h} = \hat{\alpha}_{\mathrm{id},h} + \hat{\rho}_{\mathrm{id},h}\, x_t$, fit on train+validation (direct per-horizon regressions, never iterated), requiring at least 100 observations.

Seasonal random walk. $\hat{x}_{t+h} = x_{t-((4-h) \bmod 4)}$: the forecast repeats the most recent observation falling in the same fiscal quarter, so the base sits 4, 8, 12, 16, or 20 quarters before the target, and at $h \in \{4, 8, 12, 16, 20\}$ it coincides with the no-change forecast $\hat{x}_{t+h} = x_t$. Quarterly accounting series are seasonally differenced Foster (1977), and the seasonal random walk is the standard naive expectation against which the quarterly-earnings literature measures skill Bernard and Thomas (1990).

Chronos-2. The public `amazon/chronos-2` checkpoint, zero-shot (no fine-tuning; it applies only its own inference-time normalization). For each origin the model receives the firm's raw (undeflated) reported values on a contiguous quarter grid, up to 12 observations through the origin with gaps entered as missing, which Chronos-2 handles natively, and forecasts all items jointly via its multivariate group attention. Its 21 native quantile levels $\{0.01, 0.05, 0.10, \ldots, 0.90, 0.95, 0.99\}$ are mapped into benchmark space through the origin-frozen monotone transform of §A.2. The squared-error track uses the conditional mean obtained by trapezoidal integration over the mapped quantile grid (flat tails beyond the 0.01/0.99 levels); the absolute-error track uses the mapped median; the density track scores Chronos-2 directly from the same mapped quantile grid, CRPS by the exact piecewise-linear quantile expression and interval coverage from the matching quantile pairs. No log score is reported: on flat contexts (≈0.4% of cells) the predicted quantiles collapse to zero-width intervals that admit no usable density even under a scale floor.

LLM panel. The frontier-LLM protocol and detailed prompts are documented in §C.

C LLM Benchmark

The LLM panel is evaluated on the same information set as FORMA and scored on the same grid, so differences across models reflect the forecaster rather than the information set. §C.1 documents the protocol; §C.2 reproduces the elicitation prompts.

C.1 Protocol

Inputs (tuple-sourced). History and origins are read directly from FORMA's stored test tuples, with no re-derivation of any input FORMA receives. Each prompt carries a 12-quarter reported history in original units, millions of dollars ($Q_{-11} \dots Q_0$, FORMA's `max_lookback`), with Compustat mnemonic labels and *relative* quarter indices, plus the firm's Fama–French-48 industry rendered as a plain-English sector name. The industry label gives the LLMs the same conditioning as FORMA's industry token and the baselines' dummies, and each of the 48 buckets is shared by hundreds of firms. The mnemonics are labels, not assumed knowledge: the system prompt carries a glossary giving every item's plain-English description and sign convention, plus the formulas of the derived subtotals shown as context (§C.2). No firm identifier and no absolute dates enter the prompt. Returned forecasts are standardized with the origin-frozen parameters of §A.2 before scoring. The prompts were generated from an earlier build of the panel that dropped some pre-listing and post-delisting quarters; for 87 of the 2,103 origins (4.1%) the LLMs therefore saw strictly less history than the final build gives FORMA (blank quarters in the prompt that the final build fills). Re-scoring on the 2,016 unaffected origins moves every model's R^2 by at most 0.005.

Origins and sampler. An eligible origin (firm, q_0) has q_0 in the test window 2010Q1–2024Q4, a valid scale at q_0, at least four quarters of history, and at least one future quarter to score; the eligible pool is 370,934 origins across 13,299 firms (the deflator-valid test firm-quarters of §A.1 that also pass these gates). For each sampled firm the sampler draws one variable-length block of *consecutive* eligible origins (length 4–20), calendar-balanced across 2010–2024; consecutive origins produce the overlapping-target-quarter panels that a companion analysis of forecast revisions requires. One shared, seeded origins file is reused by every model to maximize pairwise overlap. The scored run is 2,103 pinned origins across 133 firms, run in three nested stages (the first 20, then 220, then all 2,103 origins of the shared file); the two prompt arms are paired on the identical origin set.

Targets. The model forecasts the 55 primitive line items (only the firm's reported subset; structurally absent items are excluded from its list and treated as zero when reconstructing derived items). The remaining 23 of the 78 targets are reconstructed mechanically from the primitives through the accounting identities of §A.5: statement subtotals such as `atq`, `ltq`, `seqq`, and `niq`. Eighteen of the 23 appear in the prompt's glossary as labeled derived context; the remaining five (`ancq`, `wcapq`, and the three carve-outs of §A.1) are reconstructed downstream and never shown to the model. Every model is therefore scored on the same 78-target $\times$ Q_1–Q_{20} grid as FORMA. Every origin is asked for the full Q_1–Q_{20} horizon regardless of how many future quarters the firm actually has in file; unavailable horizons are masked only at scoring time, so the prompt never leaks the firm's data-end or delisting.

Two elicitation arms. Both arms share the same inputs, the same 55-primitive schema, and the same strict primitives-only output format; they differ only in one system-prompt block. The *unstructured* arm pairs the line-item glossary and accounting-consistency checks with a free-form steering paragraph that tells the model to reason about trajectory, growth, seasonality, mean-reversion, the identities, and the drivers, without prescribing a method. The *structured* arm replaces that paragraph with an explicit driver-hierarchy / roll-forward method (revenue year-over-year off the same quarter a year earlier, cost and

expense lines as ratios to revenue, working capital via turnover ratios, PP&E by roll-forward, then cash and retained earnings *set* by the cash-flow and retained-earnings rolls), using the identities as the generative construction order. The comparison therefore carries no parse-format risk. The headline table reports each model's better arm, the unstructured arm for all three models. Unstructured/structured change-space R^2 is 0.195/0.183 for Opus 4.8, 0.184/0.177 for GPT-5.5, and 0.165/0.144 for Sonnet 5; MAE is 0.361/0.367, 0.363/0.369, and 0.368/0.377.

Models and settings. The panel comprises three frontier models, Claude Opus 4.8, GPT-5.5, and Claude Sonnet 5, all with extended reasoning enabled. The Anthropic models run at their adaptive default (bounded, measured at ≈4–6k thinking tokens for Opus and ≈12–17k for Sonnet 5); GPT-5.5 runs at medium reasoning effort. Reasoning is therefore not token-matched across vendors (the batch gateway exposes no common reasoning-effort setting), a run property to bear in mind when comparing models. All calls use default temperature and `max_tokens`=64,000 (headroom for heavy reasoning on the structured prompt; billed only when consumed) and route through the Portkey batch gateway: the Anthropic models via the Messages batch endpoint, and GPT-5.5's scored run via the OpenAI Responses endpoint to capture reasoning summaries (≈82% of origins carry one). The main run is a single call per origin.

Decoding-noise subset. A 200-origin simple-random subset (drawn from the main sample) is queried $K=5$ times on every model and both arms at default temperature to measure across-call dispersion; the main run's single pass serves as draw 1, so the extra cost is four replicate runs per cell. The subset supports the companion analysis of forecast revisions and does not enter the headline results.

Common sample. The LLM column of the headline table is an exact common sample: a (firm, origin, item, horizon) cell counts only where all seven forecasters (FORMA and the six LLM runs, two prompt arms × three models) predict it finitely and a realized value is available to score against. That intersection is 2,161,399 cells, of a possible $78 \times 20 \times 2{,}103 = 3{,}280{,}680$ (targets × horizons × origins), a 66% subset; the full model suite is then re-scored inside that footprint. Parse failures are essentially absent (every model returned forecasts on ≈100% of requested cells), so exclusions come from cells with no realized value to score or outside a model's covered item set at that origin.

C.2 Exact prompts

Prompts are reproduced from the released prompt files with Unicode box-drawing and math glyphs transliterated to ASCII for typesetting; the byte-exact originals are included in the release. The `{lookback_qs}` / `{lookback_first}` / `{max_horizon}` tokens are substituted from config at load time and render as `12` / `11` / `20` for the production run.

C.2.1 System prompt, unstructured arm, in full

```
You are a financial forecasting model. Given {lookback_qs} quarters of historical
financial data (Q-{lookback_first} through Q0) for one company, forecast forward
quarters Q1 through Q{max_horizon}.

Forecasting these line items accurately is a hard, multi-step reasoning
problem. Before committing to any numbers, reason carefully and thoroughly:
study the firm's recent trajectory and growth rates, seasonality, mean-
reversion, the accounting identities that link the statements, and the likely
path of each driver. Think deeply and check your work before answering. Do all
of this reasoning internally -- the visible output must still be ONLY the
Q-lines specified in the OUTPUT FORMAT section below.

Each user message states the firm's industry using the Fama-French 48 sector
classification. Treat it as a prior -- sector-typical margins, seasonality,
```

```
asset intensity, capital structure, and growth differ markedly across
industries -- but always defer to the firm's own reported history where the two
disagree.

All values are in millions USD. Negative = losses / outflows / contra-equity.
Variable names are Compustat quarterly codes. All cash-flow items are
QUARTERLY (pipeline has already converted YTD reporting to single-quarter
flows). Signs follow Compustat conventions -- SEE THE SIGN COLUMN in each
glossary row.

================================================================================
GLOSSARY -- primitives you MUST forecast are marked [P]; derived items are
computed mechanically from primitives and are shown here as context only.
================================================================================

-- INCOME STATEMENT -------------------------------------------------------
  [P] revtq    Revenue, total                                          (+)
  [P] cogsq    Cost of goods sold                                      (+)
  [P] xsgaq    Selling, general & admin expense -- INCLUDES xrdq        (+)
  [P] xrdq     Research & development expense (informational sub of xsgaq) (+)
  [P] dpq      Depreciation & amortization expense                     (+)
  [P] stkcoq   Stock-based compensation expense (informational)        (+)
  [P] xintq    Interest expense                                        (+)
  [P] nopiq    Non-operating income/(expense), net                  (signed)
  [P] spiq     Special items (impairments, restructuring, etc)      (signed; usually <= 0)
  [P] txtq     Income tax expense                                   (signed)
  [P] miiq     Noncontrolling interest (income attrib. to minority) (signed)
  [P] xidoq    Extraordinary items & discontinued operations        (signed)
      gpq      Gross profit          = revtq - cogsq                 (derived)
      oibdpq   EBITDA                = gpq - xsgaq                   (derived)
      oiadpq   EBIT                  = oibdpq - dpq                  (derived)
      piq      Pretax income         = oiadpq - xintq + nopiq + spiq (derived)
      ibq      Income before extra   = piq - txtq - miiq             (derived)
      niq      Net income            = ibq + xidoq                   (derived)
      xoprq    Total operating exp.  = cogsq + xsgaq                 (derived)

-- BALANCE SHEET -- ASSETS ------------------------------------------------
  [P] cheq     Cash & short-term investments                           (+)
  [P] rectq    Accounts receivable, trade                              (+)
  [P] invtq    Inventories                                             (+)
  [P] acoq     Other current assets                                    (+)
  [P] ppentq   Property, plant & equipment, NET                        (+)
  [P] dpactq   Accumulated depreciation                                (+)
  [P] gdwlq    Goodwill                                                (+)
  [P] intanoq  Other intangibles (excl. goodwill)                      (+)
  [P] aoq      Other assets -- AGGREGATE that already contains intanq   (+)
      intanq   Intangibles total     = gdwlq + intanoq                 (derived)
      ppegtq   PP&E gross            = ppentq + dpactq                 (derived)
      actq     Current assets total  = cheq + rectq + invtq + acoq     (derived)
      atq      Total assets          = actq + ppentq + aoq             (derived)

-- BALANCE SHEET -- LIABILITIES -------------------------------------------
  [P] apq      Accounts payable, trade                                 (+)
  [P] dlcq     Debt in current liabilities (short-term debt)           (+)
  [P] txpq     Income taxes payable                                    (+)
  [P] drcq     Deferred revenue, current                               (+)
  [P] lcoq     Other current liabilities                               (+)
  [P] drltq    Deferred revenue, long-term -- subset of loq             (+)
  [P] dlttq    Long-term debt                                          (+)
  [P] txditcq  Deferred taxes & investment tax credit                (signed; usually +)
  [P] loq      Other liabilities -- AGGREGATE that already contains drltq (+)
      lctq     Current liab. total   = apq + dlcq + txpq + lcoq        (derived)
                                       (drcq is typically inside lcoq;
                                        do not add drcq again)
      ltq      Total liabilities     = lctq + dlttq + txditcq + loq    (derived)

-- BALANCE SHEET -- EQUITY ------------------------------------------------
  [P] cstkq    Common stock, par value                                 (+)
  [P] capsq    Additional paid-in capital / capital surplus            (+)
  [P] req      Retained earnings -- already includes AOCI             (signed)
  [P] acomincq AOCI component inside req (disclosure breakout)       (signed)
  [P] tstkq    Treasury stock at cost -- POSITIVE magnitude, subtracted  (+)
  [P] pstkq    Preferred stock                                         (+)
  [P] mibtq    Noncontrolling interest, total                          (+)
```

```
      ceqq     Common equity          = cstkq + capsq + req - tstkq     (derived)
      seqq     Total stockholders eq. = ceqq + pstkq                    (derived)

-- CASH FLOW (all quarterly; YTD already converted) -------------------------
Operating:
  [P] oancfq   Cash from operating activities                          (signed)
  [P] fopoq    Other operating CF adjustments                          (signed)
Investing (individual components; ivncfq is DERIVED from them):
  [P] capxq    Capital expenditures -- positive OUTFLOW                   (+)
  [P] ivchq    Increase in LT investments -- positive OUTFLOW             (+)
  [P] aqcq     Acquisitions -- positive OUTFLOW                           (+)
  [P] sivq     Sale of investments -- positive INFLOW                     (+)
  [P] sppeq    Sale of property -- positive INFLOW                        (+)
  [P] ivstchq  Change in ST investments (separate from ivchq)          (signed)
  [P] ivacoq   Other investing activities                              (signed)
      ivncfq   Cash from investing    = -capxq - ivchq - aqcq
                                        + sivq + sppeq + ivstchq + ivacoq (derived)
Financing (individual components; fincfq is DERIVED from them):
  [P] sstkq    Stock issuance -- positive INFLOW                          (+)
  [P] prstkcq  Stock repurchases -- positive OUTFLOW                      (+)
  [P] dltisq   Long-term debt issuance -- positive INFLOW                 (+)
  [P] dltrq    Long-term debt repayment -- positive OUTFLOW               (+)
  [P] dlcchq   Changes in short-term debt                              (signed)
  [P] dvq      Dividends paid -- positive OUTFLOW                         (+)
  [P] txbcofq  Excess tax benefit of stock options                     (signed; often 0 post-2016)
  [P] fiaoq    Other financing activities                              (signed)
      fincfq   Cash from financing    = sstkq - prstkcq + dltisq - dltrq
                                        + dlcchq - dvq + txbcofq + fiaoq (derived)
FX & free cash flow:
  [P] exreq    Effect of exchange rate on cash                         (signed)
      fcfq     Free cash flow         = oancfq - capxq                   (derived)

================================================================================
ACCOUNTING CONSISTENCY CHECKS (your primitives must satisfy these)
================================================================================

  (1) Accounting equation:   atq = ltq + mibtq + seqq
  (2) Cash rollforward:      cheq[t] = cheq[t-1] + oancfq + ivncfq + fincfq + exreq
  (3) Retained earnings:     req[t] ~= req[t-1] + niq - dvq + Delta acomincq
                             (approximate; non-cash AOCI + reclassifications add noise)

================================================================================
OUTPUT FORMAT
================================================================================

Each user message lists the EXACT subset of [P] items this firm reports
(items it never discloses are excluded -- they are treated as 0 by us). You
must:
  * Forecast EVERY item in that per-firm list.
  * Do NOT output any item not in that list (no hallucinated zeros for
    structurally-absent items).
  * Do NOT output any derived item -- they are recomputed mechanically.

One line per horizon, values comma-separated. Output Q1 through Q{max_horizon}
(forecast EVERY horizon in that range, even if you are uncertain a later
quarter will materialize -- do not stop early):

Q1: revtq=X.XX, cogsq=X.XX, xsgaq=X.XX, ...
Q2: revtq=X.XX, cogsq=X.XX, xsgaq=X.XX, ...
...
Q{max_horizon}: revtq=X.XX, cogsq=X.XX, xsgaq=X.XX, ...

Output ONLY the Q-line forecasts. No markdown, no commentary, no code fences.
```

C.2.2 Structured arm: the block that replaces the steering paragraph

The remainder of the prompt (industry prior, units and sign conventions, glossary, consistency checks, output format) is unchanged from the unstructured arm.

```
PROJECTION METHOD -- forecast a few DRIVERS, then DERIVE the rest. Reason
through this sequence internally (in growth rates and ratios), then convert
back to dollar levels before emitting:
```

```
  1. Revenue (revtq) first. Anchor on the SAME fiscal quarter a year earlier:
     revtq[t] ~= revtq[t-4] x (1 + g), where g is a year-over-year growth rate
     read from the recent trailing-4-quarter trend and FADED toward a modest
     long-run rate across the 20-quarter horizon -- do not let a recent spike
     persist to Q20. Anchoring on t-4 preserves seasonality.

  2. Operating lines as ratios to revenue. Project cogsq, xsgaq (which
     includes xrdq), dpq, and stkcoq each as a share of revtq, held near its
     trailing-4-quarter median and drifting only with a clear operating-
     leverage reason; convert back to a level = ratio x revtq[t]. Income tax
     (txtq) ~= tax rate x pretax income, and tracks the sign of pretax income.

  3. Working capital via turnover. Project receivables (rectq) off revtq
     (days-sales-outstanding) and inventory (invtq) and payables (apq) off
     cogsq (days-inventory / days-payable); hold each turnover near its
     trailing level, then convert back to a level using projected revtq/cogsq.

  4. PP&E by roll-forward. ppentq[t] ~= ppentq[t-1] + capxq - dpq, with capxq
     projected as a share of revtq.

  5. Cash flow first, then SET cash and retained earnings by roll-forward --
     do NOT forecast cheq or req independently:
       * Operating cash flow must be consistent with earnings and working
         capital: oancfq ~= niq + dpq - Delta rectq - Delta invtq + Delta apq. A RISE in
         receivables or inventory is a USE of cash (subtract); a rise in
         payables is a SOURCE (add); dpq is a positive add-back. capxq is a
         cash OUTFLOW (in ivncfq); debt issuance is a source while repayments
         and dividends (dvq) are uses (in fincfq).
       * Cash: cheq[t] = cheq[t-1] + oancfq + ivncfq + fincfq + exreq. Set
         cheq to this rollforward result, not to an independent guess.
       * Retained earnings: req[t] ~= req[t-1] + niq - dvq + Delta acomincq. Set req
         to this roll.

Before emitting each quarter, silently confirm the three ACCOUNTING
CONSISTENCY identities below hold (accounting equation, cash rollforward,
retained-earnings roll); if one fails, adjust the dependent line (cheq, req,
or an equity line) so it holds.

Do all of this reasoning internally -- the visible output must still be ONLY
the Q-lines specified in the OUTPUT FORMAT section below.
```

C.2.3 Per-origin user message (both arms)

Assembled per origin: the fixed horizon instruction, the firm's Fama–French-48 industry, the firm's active-primitive list, then the 12-quarter history as CSV-like blocks (one per statement; rows null across all history are suppressed). Schematic, with illustrative values and the history truncated to 3 of 12 quarters for space:

```
Forecast horizons: Q1 through Q20.

Firm industry (Fama-French 48): Retail.

This firm reports 47 primitive line items. Forecast ONLY these
items -- do not output any other primitives:

revtq, cogsq, xsgaq, dpq, xintq, ..., oancfq, capxq, dvq

Items not in this list are structurally absent for this firm
(consistently null in history); they will be treated as 0.

Historical data:

=== INCOME STATEMENT ===
Account,Q-11,...,Q-1,Q0
```

```
revtq,1203.40,...,1456.10,1502.77
cogsq,742.10,...,889.30,910.55
...
=== BALANCE SHEET - ASSETS ===
Account,Q-11,...,Q-1,Q0
cheq,310.22,...,402.10,419.88
...
```

The model returns one comma-separated line per horizon (`Q1: revtq=..., cogsq=..., ...` → `Q20: ...`), primitives only; the 23 derived and subtotal items are reconstructed mechanically downstream.

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